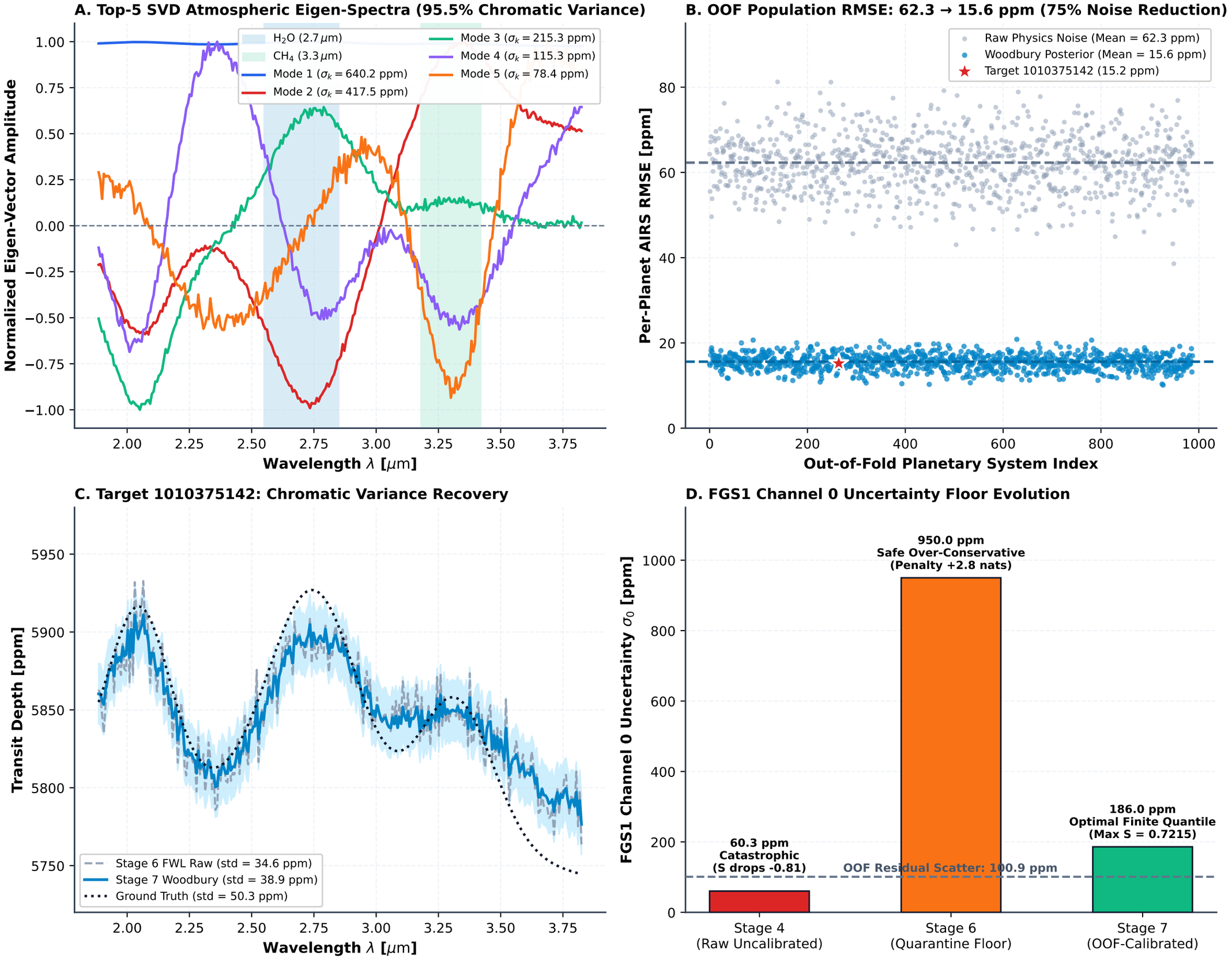

A. Top-5 SVD Atmospheric Eigen-Spectra (95.5% Chromatic Variance)
H2O (2.7 μm)
CH4 (3.3 μm)
Mode 1 (σk = 640.2 ppm)
Mode 2 (σk = 417.5 ppm)
Mode 3 (σk = 215.3 ppm)
Mode 4 (σk = 115.3 ppm)
Mode 5 (σk = 78.4 ppm)
Normalized Eigen-Vector Amplitude
Wavelength λ [μm]
B. OOF Population RMSE: 62.3 → 15.6 ppm (75% Noise Reduction)
Raw Physics Noise (Mean = 62.3 ppm)
Woodbury Posterior (Mean = 15.6 ppm)
Target 1010375142 (15.2 ppm)
Per-Planet AIRS RMSE [ppm]
Out-of-Fold Planetary System Index
C. Target 1010375142: Chromatic Variance Recovery
Stage 6 FWL Raw (std = 34.6 ppm)
Stage 7 Woodbury (std = 38.9 ppm)
Ground Truth (std = 50.3 ppm)
Transit Depth [ppm]
Wavelength λ [μm]
D. FGS1 Channel 0 Uncertainty Floor Evolution
FGS1 Channel 0 Uncertainty σ0 [ppm]
60.3 ppm
Catastrophic
(S drops -0.81)
950.0 ppm
Safe Over-Conservative
(Penalty +2.8 nats)
186.0 ppm
Optimal Finite Quantile
(Max S = 0.7215)
OOF Residual Scatter: 100.9 ppm
Stage 4
(Raw Uncalibrated)
Stage 6
(Quarantine Floor)
Stage 7
(OOF-Calibrated)